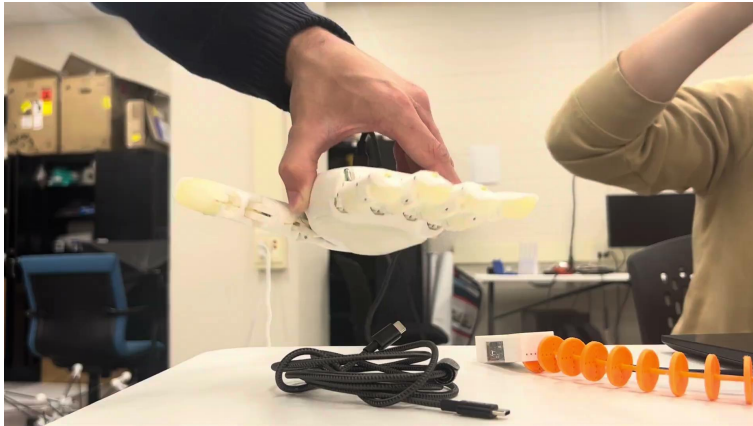


Success & Shortcomings of fixed finger



Issues Evident From CI reports

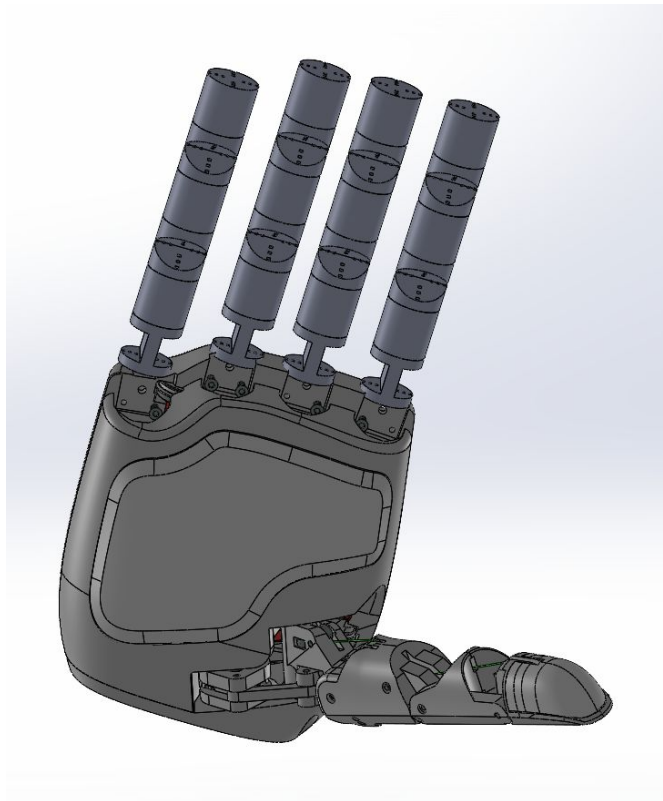
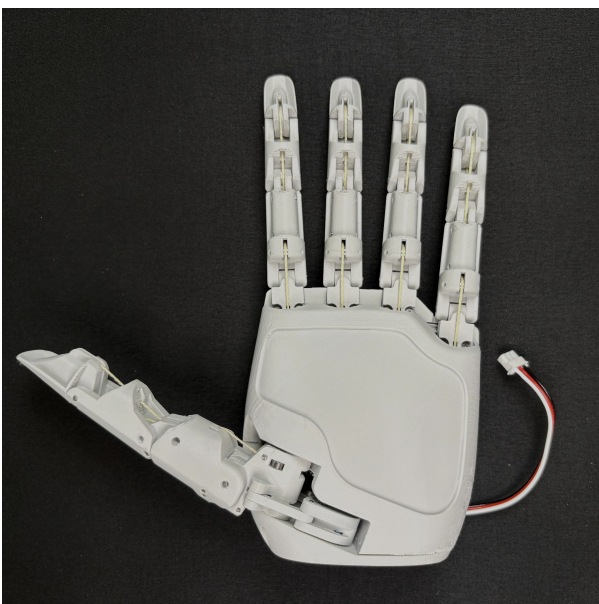
“Cylindrical and irregularly shaped objects remain the hardest challenge”

“ egg is a small object...egg will not contact the silicon grip at all

“Attempts to grasp failed” (pick up a screwdriver /w long side parallel to table)

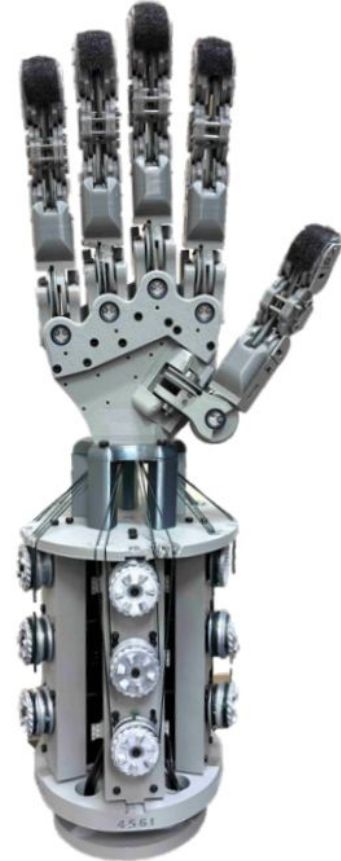
“Wide & thin objects do not work”

“Small objects do not work”

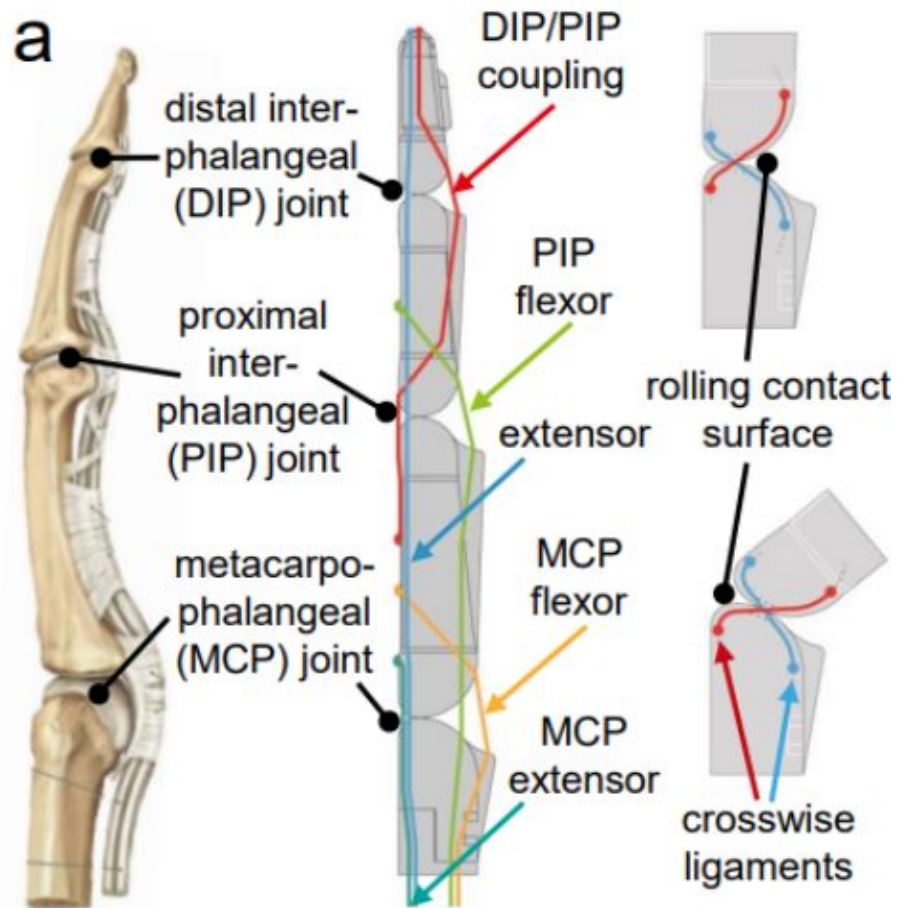


TetherIA, "Aero_Hand_Open_Right.stp," *aero-hand-open*, GitHub repository, hardware/CAD/, 2025. [Online]. Available: https://github.com/TetherIA/aero-hand-open/blob/main/hardware/CAD/Aero_Hand_Open_Right.stp. [Accessed: May 28, 2026].

A



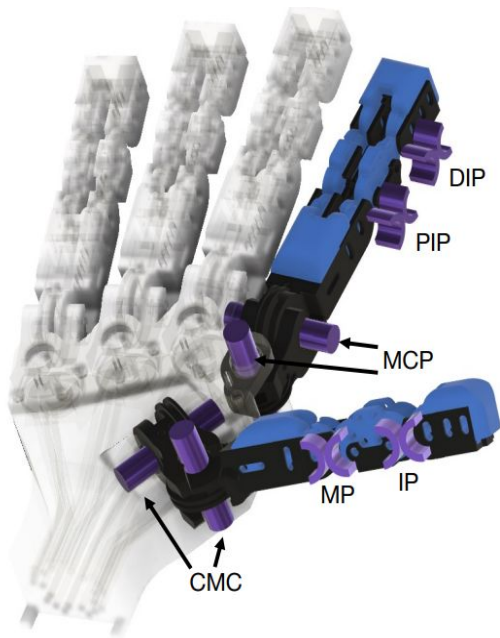
L. Lin, S. Patel, J. Moon, S. Lazebnik, and U. Jain, "CRAFT: A Tendon-Driven Hand with Hybrid Hard-Soft Compliance," *arXiv preprint arXiv:2603.12120*, 2026, Fig. 1.



Printed hand with tendons and ligaments. (Image Credit: ETH Zurich)

Grasping Mechanics For Robot Fingers

33/33 Standard grasp types (Yale GRASP)

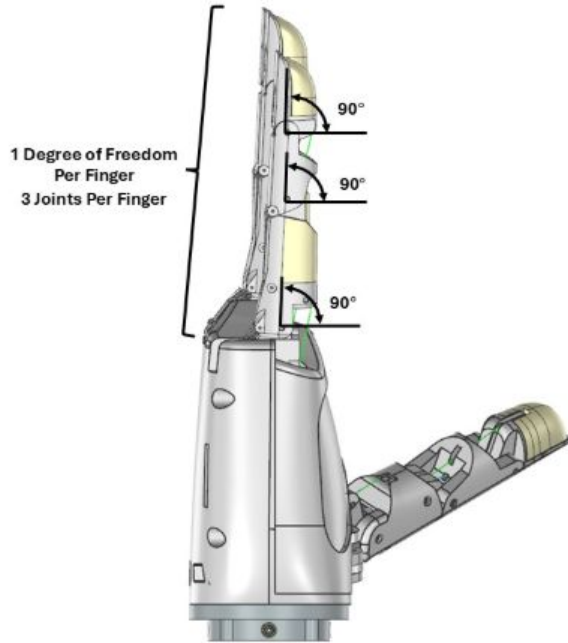


Opposition Type: Virtual Finger 2:	Power						Intermediate			Precision				
	Palm		Pad				Side			Pad				Side
	3-5	2-5	2	2-3	2-4	2-5	2	3	3-4	2	2-3	2-4	2-5	3
Thumb Abd.														
Thumb Add.														

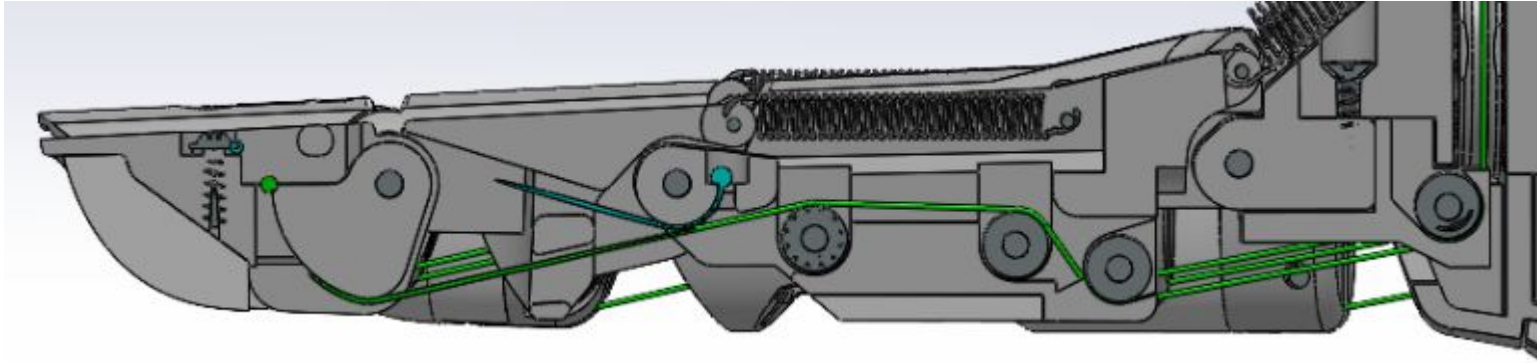
Requires a 90° flex on (MCP, PIP, DIP joints) & abduction [Shaw et al., Demonstrating LEAP Hand v2; Wonik Robotics, Allegro Hand]

Passive Compliance: to best grasp objects, certain segments need to be able to conform slightly to the item. (Odhner et al., Deimel and Brock).

Review of Popular Tendon-Driven Dexterous Fingers: TetherIA Aero Hand



Review of Popular Tendon-Driven Dexterous Fingers: TetherIA Aero Hand

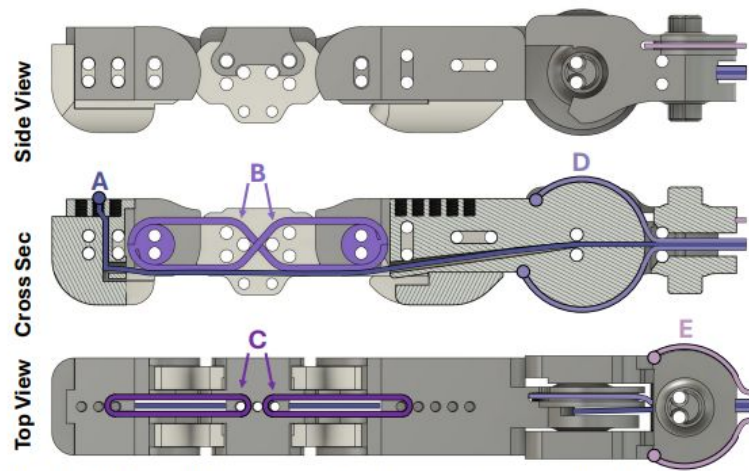


Aero Hand uses 1 tendon, a series of hinges, and 2 springs to introduce compliance and grasping mechanics.

When a force from a stagnant object is introduced the tendon forces actuation on all joints above beginning with joint furthest from object

When no large object is present first the MCP flexes then the DIP and lastly the PIP (knuckle, fingertip)

Review of Popular Tendon-Driven Dexterous Fingers: CRAFT HAnd



3 Actuators/ Tendons Per Finger (One Controls left/right, one Controls MCP, One controls PIP & DIP).

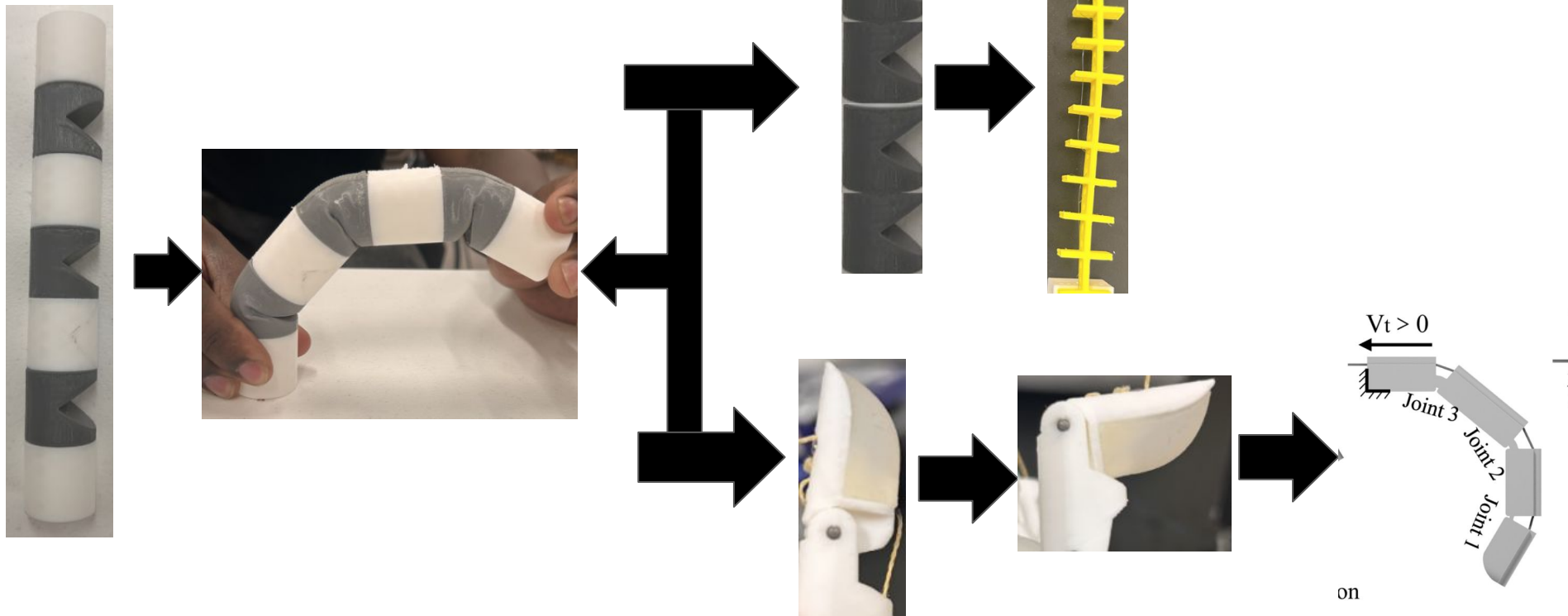
Highly Dexterous, Abduction originates from soft joint padding on L/R MCP

Takeaways from Popular hands:

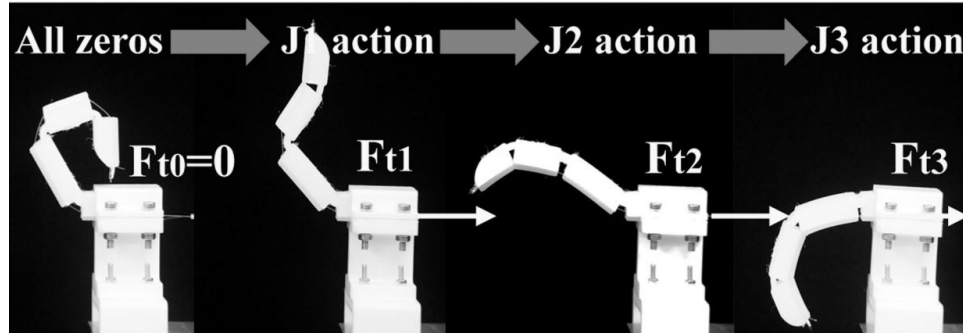
Reliance on soft joint, rigid “skeleton” for dextrous action.

Standard grip w/o abduction is possible for one tendon with careful design of materials & joint structure

Applications on Current design

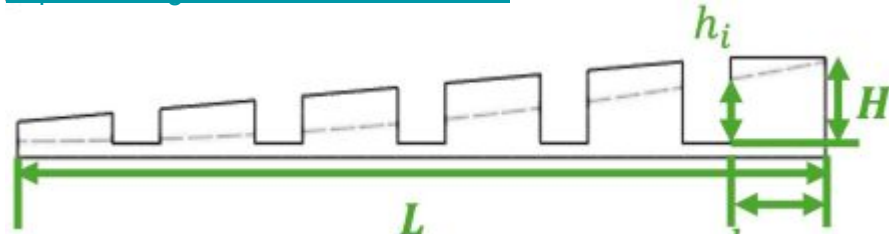


Design Language of All-Soft Tendon-Driven Fingers



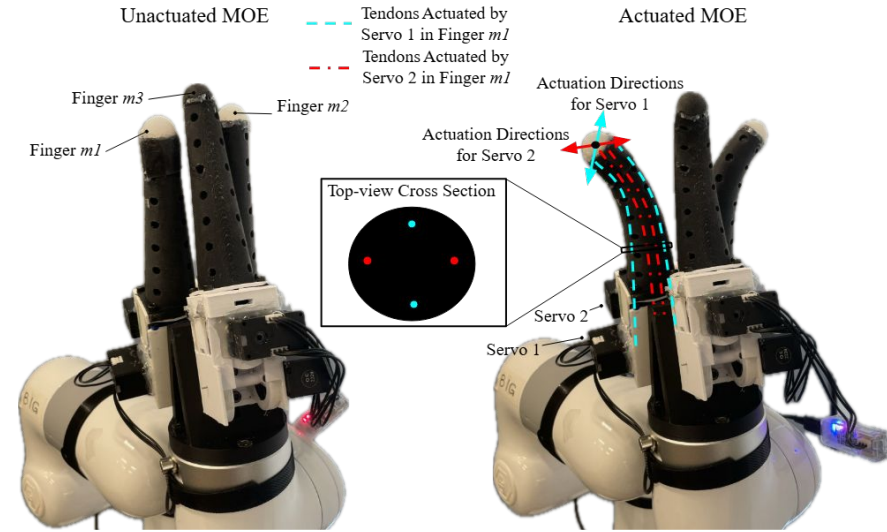
Zhang, Y., Zhang, W., Yang, J., & Pu, W. (2022). Bioinspired soft robotic fingers with sequential motion based on tendon-driven mechanisms. *Soft Robotics*, 9(3), 531-541.

<https://doi.org/10.1089/soro.2021.0009>



Yi, S., Bai, X., Singh, A., Ye, J., Tolley, M. T., & Wang, X. (2025). Co-design of soft gripper with neural physics. *arXiv*.

<https://doi.org/10.48550/arxiv.2505.20404> (Note: This paper was also accepted as a poster presentation at the 2025 Conference on Robot Learning [CoRL]).



Soft Robotic Dynamic In-Hand Pen Spinning (CMU)

The Soft Approach: Single-Tendon Grasping Soft Fingers

Issues:

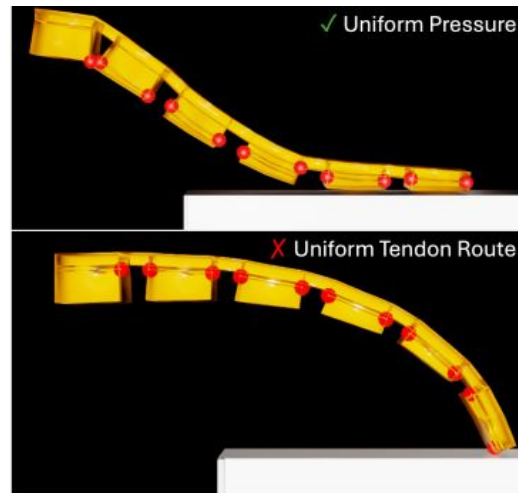
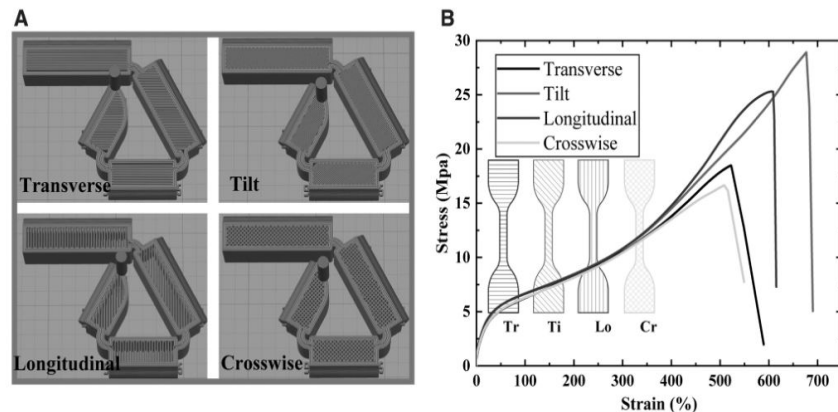
Single tendon, causes finger to curve unevenly in an arc, not grasping around object

solution :

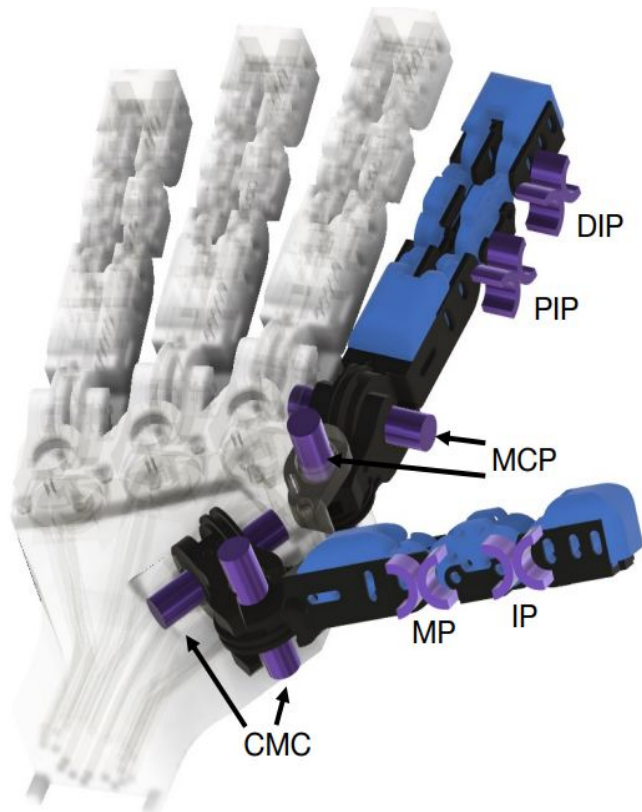
Take design inspiration from “Co-Design of Soft Gripper with Neural Physics “ and “Bioinspired Soft Robotic Fingers with Sequential Motion Based on Tendon-Driven Mechanisms”

Implementation:

Different infill rates and printing directions alter the behavior of “skeletal” and “joint” finger portions. & use diagonal tendon placement alleged with earlier formula to get even pressure



Terminology Refresher



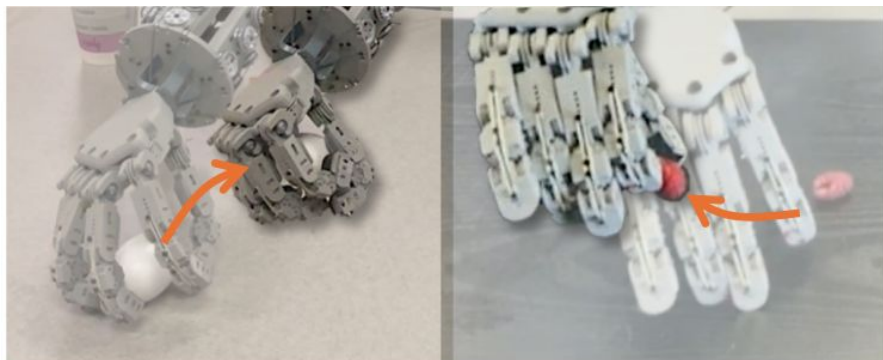
33/33 Standard grasp types (Yale GRASP)

Opposition Type:
Virtual Finger 2:

	Power						Intermediate			Precision				
	Palm		Pad				Side			Pad				Side
	3-5	2-5	2	2-3	2-4	2-5	2	3	3-4	2	2-3	2-4	2-5	3
Thumb Abd.														
Thumb Add.														

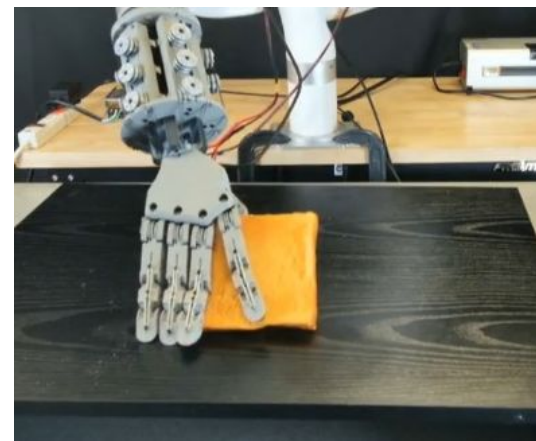
Diagram of CRAFT hand Showing joint names

What Can the CRAFT Finger Do?: (basically everything)



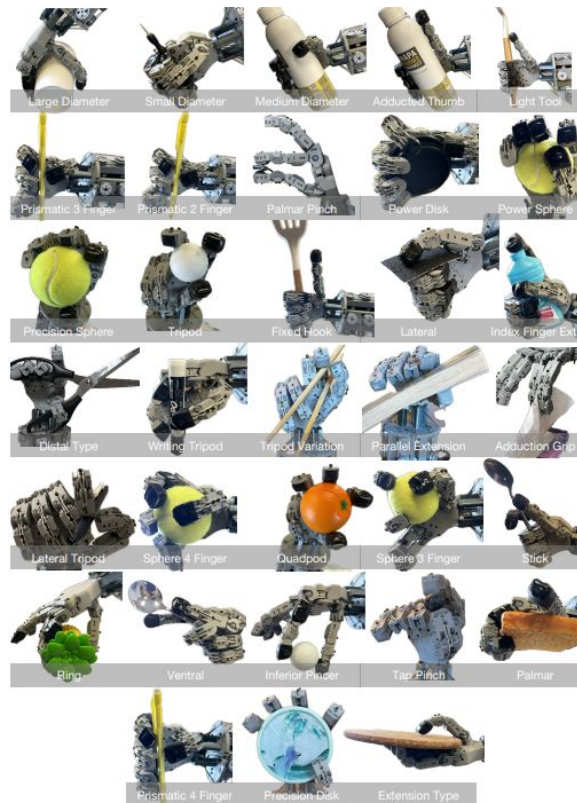
Pick Up Egg

Pick Up Raspberry



Small, round, fragile objects (TetherIA Couldn't)

Wide & flat objects



33/33 Distinct
“Human”
grasps

(as identified by
yale’s GRASP
Lab)

Advantages of CRAFT Finger

Separate actuation on the MCP and PIP/DIP joints allows for tasks like picking small objects from tables

Motion of DIP & PIP is synchronised and connected (this is exactly how our fingers work so it allows for all human movement of these joints)

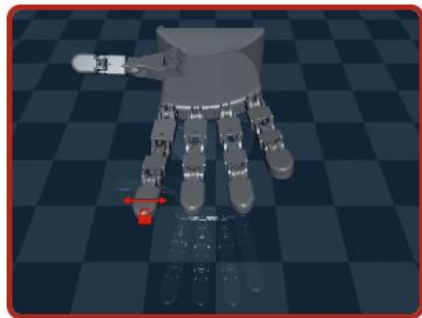
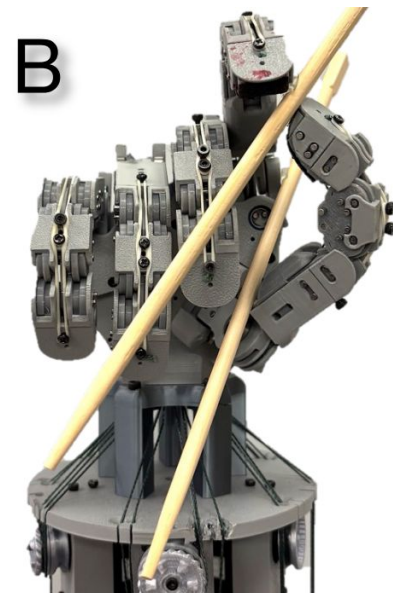


The Case for Abduction

When CRAFT (an abducted hand) and RUKA (non-abducted hand) were compared, The primary benefit of abduction is the “Ability to manipulate objects in-hand, such as pens or chopsticks” (Lin 5).

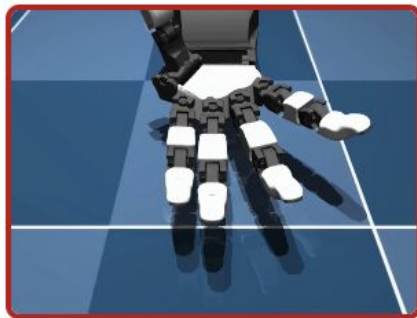
experiments on Allegro & LEAP showed a hand must periodically "let go" of an object to re-orient it

Abduction of the MCP joint is required to meet the last 5 of the 33 standard human Grasps (Yale GRASP)



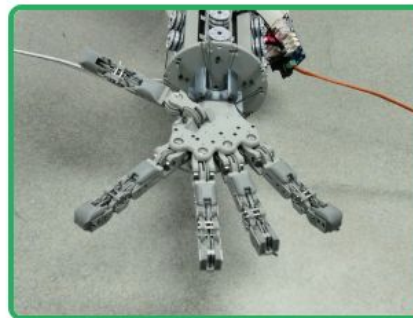
RUKA

✗ Hold pen or chopsticks



ORCA

✗ Power grip small tools
Pick up coins



CRAFT

✓ Hold pen or chopsticks
Power grip small tools
Pick up coins

Lin, L., Patel, S., Moon, J., Lazebnik, S., & Jain, U. (2024). CRAFT: A Tendon-Driven Hand with Hybrid Hard-Soft Compliance.

Feix, T., Romero, J., Schmiedmayer, H. B., Dollar, A. M., & Kragic, D. (2015). The grasp taxonomy of human grasp types. IEEE Transactions on Human-Machine Systems, 46(1), 66-77.

Alternative Solutions

RUKA got 28/33 standard human grasps
~88% WITHOUT ABDUCTION

How? - Passive Splay

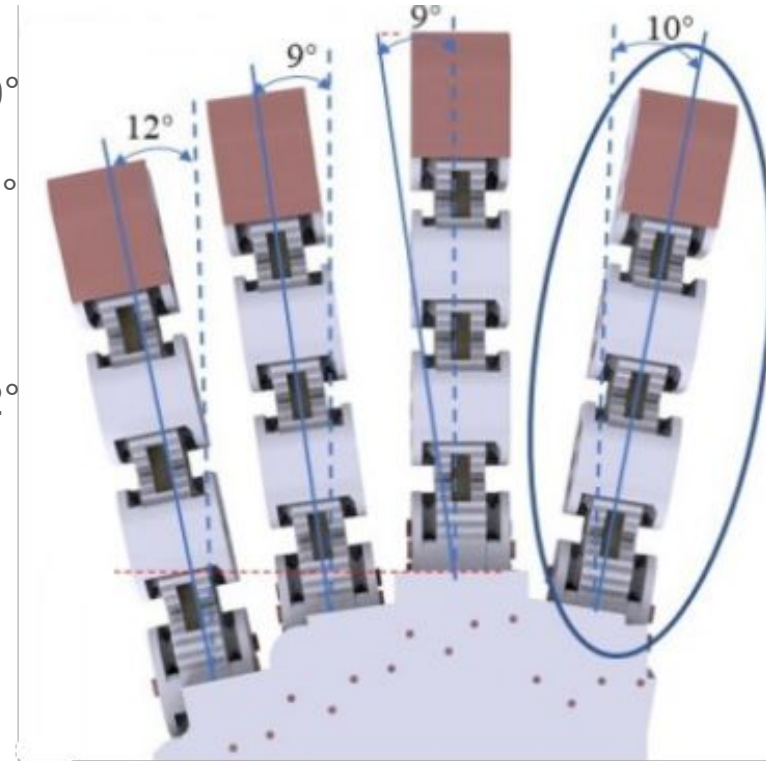
Each finger is offset by x degrees from the vertical "forearm"

When hand flat, fingers spread apart
wide maximizes gripping space, but
As tendon pulls, fingertips converge
toward the center of the palm

mimics the natural human grasping
(fingers spread apart to open and pull
together to grab)

BPI softhand
optimized
simulation to find
ideal angle:

Index Finger: 10°
outward,
Middle Finger: 0°
(straight)
Ring Finger: 9°
outward
Pinky Finger: 12°
outward



Ideas On My hand

Only use 3 fingers, why?

- Standard grasp Dataset shows that (PIP) joints of the ring and little finger move together 71% of the time, and MCP move together 76.3%
- SABD hand (combined 4th + 5th) achieved an 86% success rate on standard grasping tests (~same as RUKA)

But why want it?

- less motors, less failure points, more reliable control + programming, (bonus easier to make modular)

